

List of questions for IA on Applications of Electromagnetics:

Write answers with neat diagrams and equations (whenever applicable)

1. What is the Hall Effect? Derive the expression for Hall voltage and mention its applications.

2. Explain the principle and working of electromagnetic pumps.

3. What is electromagnetic interference (EMI)? Discuss its

Show more ▾

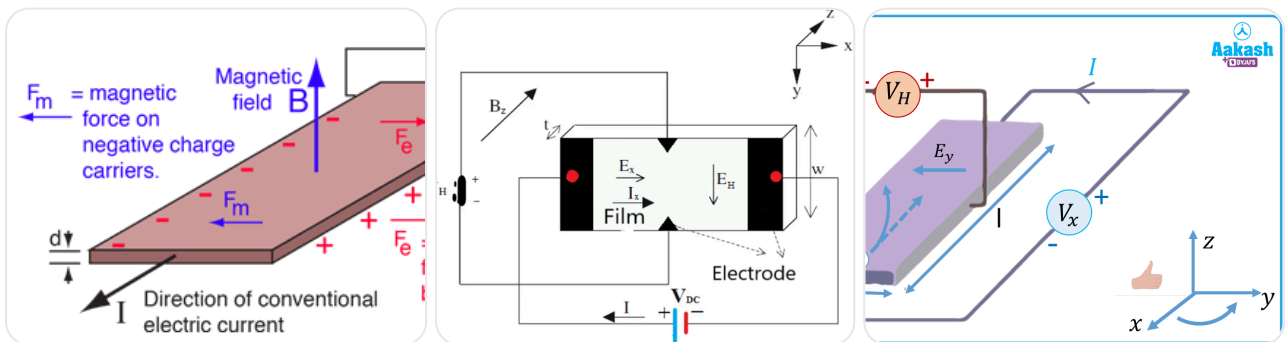
Here are clear, exam-ready answers with diagrams and equations for your IA:

1. Hall Effect

Definition

The **Hall Effect** is the phenomenon in which a voltage (Hall voltage) is developed across a current-carrying conductor or semiconductor when it is placed in a magnetic field perpendicular to the current.

Diagram



Derivation of Hall Voltage

Consider:

- Current I flowing along x-axis
- Magnetic field B along z-axis
- Charge carriers deflect along y-axis

Step 1: Magnetic Force

$$F = qv_d B$$

Where:

- q = charge
 - v_d = drift velocity
-

Step 2: Electric Force (due to Hall field E_H)

$$F = qE_H$$

Step 3: At equilibrium

$$qE_H = qv_d B \Rightarrow E_H = v_d B$$

Step 4: Hall Voltage

$$V_H = E_H \cdot w$$

$$V_H = v_d B w$$

Now,

$$v_d = \frac{I}{nqA}$$

Substitute:

$$V_H = \frac{IB}{nqt}$$

Final Expression

$$V_H = \frac{IB}{nqt}$$

Where:

- n = charge carrier density
- t = thickness of conductor

Applications

- Measurement of magnetic field (Hall probe)
- Determination of carrier concentration
- Identification of semiconductor type (n-type / p-type)
- Used in sensors (position, speed, current)
- Brushless DC motors

2. Electromagnetic Pumps

Principle

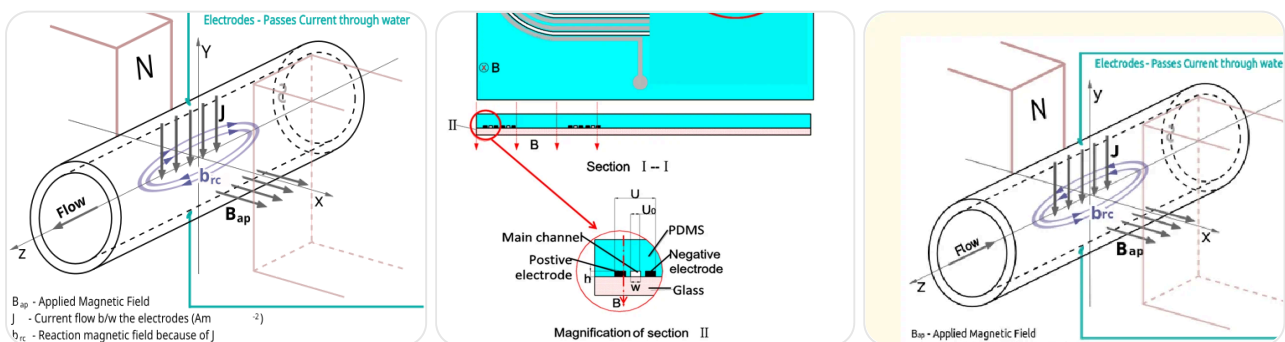
Electromagnetic pumps work based on **Lorentz force**:

$$\vec{F} = q(\vec{v} \times \vec{B})$$

For a conductor carrying current:

$$F = BIL$$

Diagram



Working

1. A conductive fluid (like liquid metal) flows through a channel

2. Electric current is passed across the fluid
 3. A magnetic field is applied perpendicular to current
 4. Due to Lorentz force, fluid experiences force and moves
-

Key Points

- No moving mechanical parts
 - Flow controlled electrically
 - Works best with conductive fluids
-

Applications

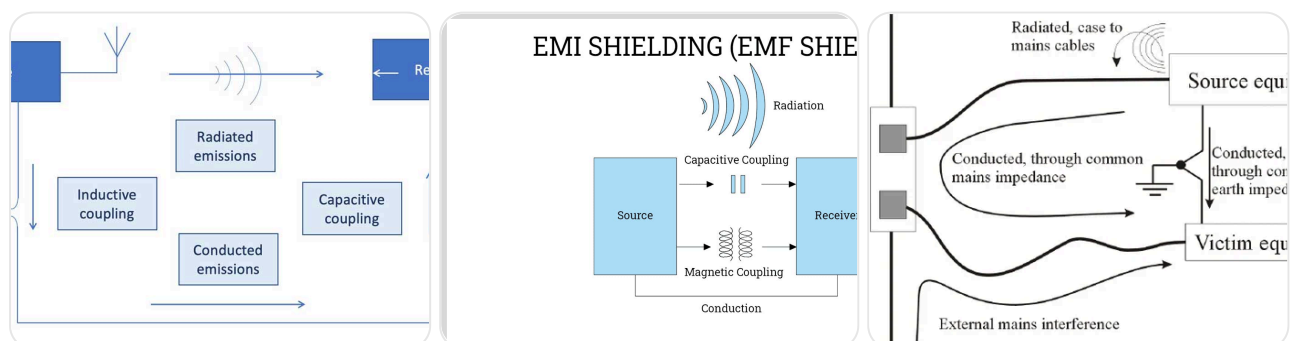
- Nuclear reactors (liquid metal cooling)
 - Metallurgy (molten metals)
 - Space propulsion systems
 - Chemical processing industries
-

3. Electromagnetic Interference (EMI)

Definition

Electromagnetic Interference (EMI) is the disturbance caused by electromagnetic radiation that affects the performance of electronic devices.

Diagram



Causes of EMI

- Switching devices (SMPS, digital circuits)
- Power lines and high-frequency signals

- Wireless transmitters (mobile phones, Wi-Fi)
 - Lightning and natural disturbances
 - Poor grounding and shielding
-

Types of EMI

- **Conducted EMI** → through wires
 - **Radiated EMI** → through air
-

Effects of EMI

- Signal distortion
 - Data loss
 - Malfunction of devices
 - Reduced system performance
 - Noise in communication systems
-

Methods to Reduce EMI

1. Shielding

- Use metal enclosures (Faraday cage)

2. Grounding

- Proper earthing reduces noise

3. Filtering

- Use capacitors, inductors (EMI filters)

4. Twisted Pair Cables

- Reduces magnetic coupling

5. Proper PCB Design

- Short traces
- Separation of analog and digital circuits

6. Ferrite Beads

- Suppress high-frequency noise
-

Applications / Importance

- Essential in communication systems
 - Used in medical electronics (ECG, MRI)
 - Important for EMI/EMC compliance
-

If you want, I can **convert this into a neat PDF**, or **add more detailed derivations/extra diagrams for marks boost** 👍



Thanks

You're welcome!

Good luck for your IA 👍

